

Photoinduced Phase Segregation in Mixed Halide Perovskites: Mechanisms, Suppression Strategies, and Device Performance Optimization

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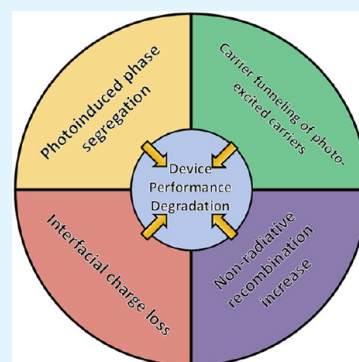
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ABSTRACT: For mixed-halide perovskite solar cells, the phenomenon of photoinduced phase segregation (PIPS) and its derivative effect of uneven spatial ion distribution within the material have induced a series of adverse consequences. Since the discovery of this phenomenon, although substantial research progress has yielded significant achievements in this field, a systematic logical organization among different research methodologies or models of PIPS remains scarce. To address this gap, this review systematically traces the evolution of PIPS research—from fundamental thin-film properties to its impact on the operational performance of complete solar cells. Through synthesizing prevalent models and corresponding solutions, it endeavors not only to summarize the intrinsic logic within each model framework but also to rigorously analyze the interrelationships among distinct models. Simultaneously, diverging from the substantial body of research focused on analyzing PIPS-affected thin films, this work systematically examines and integrates the implications of the PIPS phenomenon within complete solar cell devices and the corresponding suppression strategies implemented at the device level. This perspective bridges critical mechanistic gaps identified in current experimental investigations while underscoring the imperative for co-optimizing phase stability and defect passivation in next-generation perovskite photovoltaics.

KEYWORDS: photoinduced phase segregation, mixed halide perovskites, device performance optimization, phase stability optimization, multi-dimensional synergy effect



1. INTRODUCTION

Globally, with the accelerating transition toward green and clean energy systems, solar cells have garnered significant scientific attention. Among various photovoltaic technologies, perovskite solar cells (PSCs) have emerged as a frontrunner in next-generation photovoltaics due to their exceptional performance characteristics. These devices demonstrate distinctive advantages in three critical aspects of photovoltaic technology: power conversion efficiency, manufacturing cost, and fabrication processes. Compared to commercial thin-film counterparts such as CIGS (23.3%) and CdTe (23.1%),¹ perovskite-based photovoltaics have achieved remarkable progress in efficiency metrics. Since their initial reported efficiency of 3.8% in 2009,² state-of-the-art single-junction PSCs now boast a certified efficiency of 27.0%¹—representing a 7.1-fold enhancement achieved through optimized fabrication formulations, refined processing techniques, and architectural innovations.

However, early experiments revealed that perovskite materials exhibit anomalous phenomena in open circuit voltage (V_{OC})³ and external quantum efficiency⁴ that deviate from conventional patterns. As researchers delved deeper into the causes of these anomalies, phase segregation emerged as a key

focus. Hoke et al.⁵ observed phase segregation in experiments and suggested that it might be related to V_{OC} losses. This marked the beginning of extensive research into PIPS, which has since been recognized as one of the factors limiting the core conversion efficiency of mixed halide perovskite materials. Consequently, the mechanisms underlying PIPS and its impact on performance have become hot topics in research. Phase segregation has been discussed in multiple reviews, covering aspects such as characterization principles and methods,⁶ various phase segregation models,^{6,7} and the effects of phase segregation on solar cell devices.⁸ Unlike previous reviews, this review emphasizes the research logic in the discussion of different phase segregation models, summarizes these models, and proposes future research directions. Additionally, it introduces new findings on the impact of phase segregation

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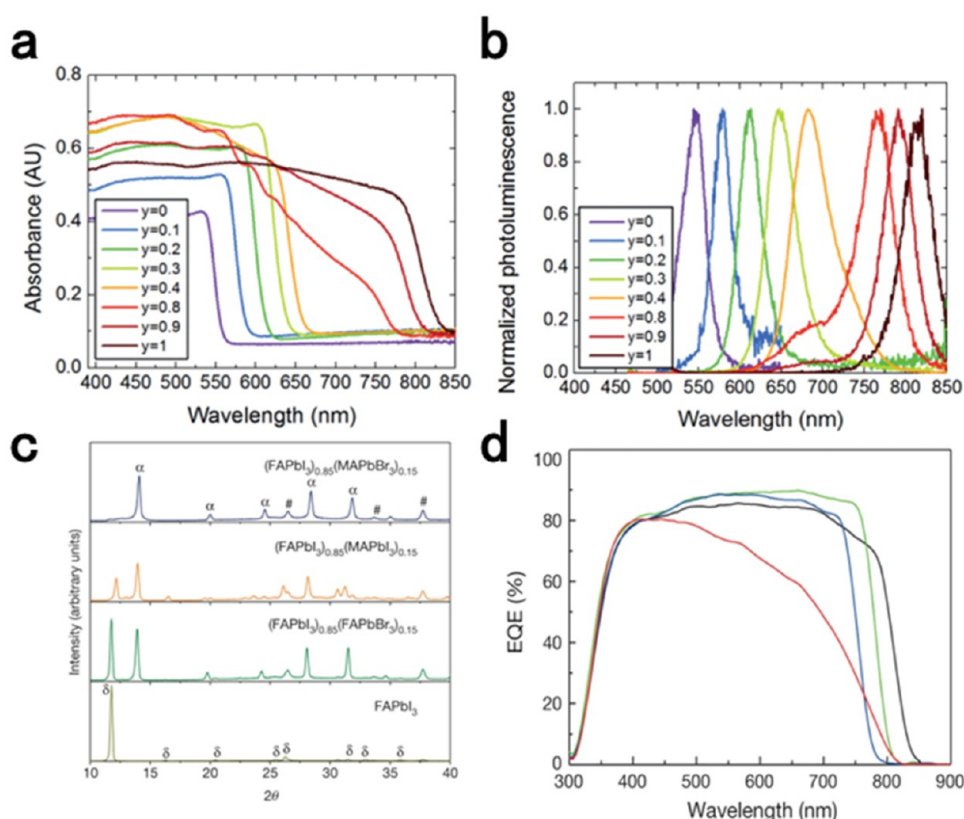


Figure 1. Influence of changes in the ratio of iodine to bromine at the X site (a). UV–vis absorbance of the FAPbI₃Br_{3y} perovskites with varying y , measured in an integrating sphere, shows the trend of UV–vis absorption spectra changing with the ratio of iodide to bromide.⁹ (b). Corresponding steady-state photoluminescence spectra for the same films, shows the trend of photoluminescence spectra changing with the ratio of iodide to bromide.⁹ Reproduced with permission from ref 9. Copyright 2014 Royal Society of Chemistry. (c). The trend of stabilization changes with the ratio of iodide to bromide.¹⁰ (d). The trend of external quantum efficiency changing with the ratio of iodide to bromide. EQE spectra for cells using (FAPbI₃)_{1-x}(MAPbBr₃)_x materials (with $x = 0$, black; $x = 0.05$, red; $x = 0.15$, green; $x = 0.25$, blue). Pure FAPbI₃ ($x = 0$) was annealed at 150 °C.¹⁰ Reproduced with permission from ref 10. Copyright 2015 Springer Nature.

on solar cells devices and summarizes methods to suppress phase segregation in these devices.

Therefore, this review will provide an in-depth exploration of PIPS in perovskite materials, starting from their composition and structure, and analyzing the phenomenon in perovskite thin films and solar cell devices in detail. It will also combine some latest research progress across different fields, analyze current challenges, and offer insights into future developments. The aim is to comprehensively present the current state of research on PIPS in perovskite materials and its potential applications across various fields, highlighting its significance for optimizing the performance of perovskite materials.

2. COMPOSITION, STRUCTURE, AND BASIC PROPERTIES OF PEROVSKITE MATERIALS

The simple perovskite structure is represented by the formula ABX₃, where A is typically a large-radius ion such as alkaline earth or rare-earth metal ions; B is a small transition metal ion; and X is an anion. Structurally, this composite metal oxide adopts a cubic crystal system. The A-site ions exhibit 12-coordination, residing in cavities formed by interconnected octahedra. The B-site ions form octahedral coordination with six X-site ions, and the X-site octahedra share vertices to form a three-dimensional network. The double perovskite structure follows the general formula A₂BB'O₆, where A is usually an alkaline earth metal, and B/B' are transition metals. These structures typically exhibit cubic or octahedral morphologies with luster and colors of corresponding perovskite thin film ranging from light to brown. Adjusting the types and ratios of B and B' site metal ions enables the

design of materials with diverse electronic and ionic conductive properties for tailored functionalities. Layered perovskite structures consist of alternating inorganic and organic layers. The inorganic layers, composed of metal and oxygen ions, resemble the perovskite crystal structure, while the organic layers, formed by organic molecules or ions, bridge the inorganic layers. This alternating arrangement enhances structural stability and tunability.

With the advent of renewable energy technologies, PSCs have attracted considerable research interest due to their cost-effectiveness, solution processability, and high efficiency. Among these, pure-iodide (e.g., MAPbI₃) and pure-bromide (e.g., MAPbBr₃) perovskite systems have emerged as focal points, though inherent limitations persist in practical applications. Pure-iodide perovskites exhibit a bandgap of ~1.55 eV, primarily absorbing visible light while underutilizing near-infrared photons (>800 nm). Conversely, pure-bromide counterparts with wider bandgaps (~2.3 eV) suffer from excessive energy losses due to restricted absorption to short-wavelength light. These limitations prompted investigations into mixed-halide perovskite systems as a strategic solution to optimize spectral utilization and advance PSC development.

In 2014, seminal work demonstrated bandgap tunability of perovskite materials through iodine–bromine ratio modulation, achieving adjustable bandgaps spanning 1.48–2.23 eV. As shown in Figure 1a,b, the changes of absorption spectra and steady-state photoluminescence spectra reflect the adjustment of band gap. This breakthrough proposed that halide-engineered perovskites could serve as top subcells in tandem architectures, where bandgap matching between subcells enhances overall efficiency.⁹ Subsequent studies introduced gradient halide (Br/I) distribution strategies to simultaneously optimize photon harvesting and charge transport. As shown

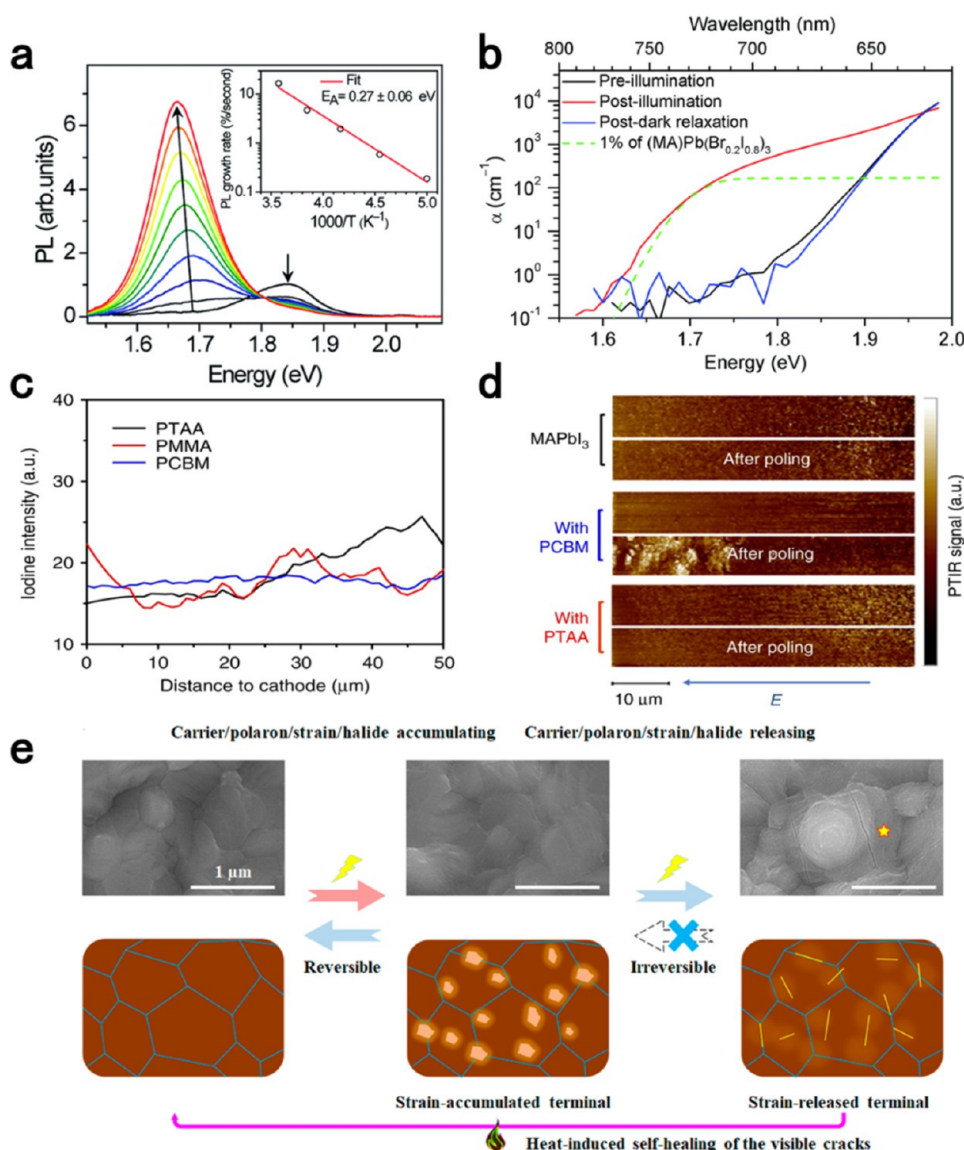


Figure 2. Experimental characterization of photoinduced phase segregation and self-healing processes in mixed-halide perovskite films: (a). PL spectrum analysis of a film at 300 K using a 457 nm, 15 mW cm⁻² light source over 45 s in 5 s increments (inset shows the temperature dependence of PL growth rate).⁵ (b). Absorption spectra of the film after 5 min of illumination at 100 mW cm⁻² (black) and after 1 h in the dark (blue), with the green curve showing the scaled absorption spectrum for $x = 0.2$.⁵ Reproduced from ref 5. Available under a [Creative Commons Attribution 3.0 Unported] license. Copyright 2014 Royal Society of Chemistry. (c). The PTAA layer, acting as a hole transport layer (HTL), can extract photogenerated holes, leaving a large number of excess electrons. These excess electrons significantly promote the migration of iodide ions. The PMMA layer serves as an insulating layer and has no obvious effect on charge extraction. The migration degree of iodide ions is intermediate.²⁶ (d). The photothermal-induced resonance (PTIR) microscope was used to show that excessive charge accelerates the migration of iodide ions (I⁻) and methylammonium cations (MA⁺) in the perovskite thin film.²⁶ Reproduced from ref 26. Available under a [Creative Commons Attribution 4.0 International] license. Copyright 2018 Springer Nature. (e). Scanning electron microscopy images and corresponding schematic diagrams of strain accumulation/release-induced PIHS/photoinduced self-healing processes.³¹ Reproduced from ref 31. Copyright 2021 American Chemical Society.

in Figure 1c,d, the changes of stabilization and external quantum efficiency both vary with the change Br/I ratio.¹⁰ Concurrently, chlorine incorporation emerged as an alternative approach, with studies revealing that minimal Cl⁻ doping passivates grain boundary defects and suppresses ion migration, thereby improving film stability. Although Cl doping showed no direct bandgap modulation effect, its contribution to performance enhancement through stability improvements was empirically validated.

Beyond halide engineering, researchers explored alternative bandgap tuning strategies through cation manipulation and process optimization. Cesium (Cs⁺) incorporation was found to enhance lattice rigidity while marginally widening bandgaps.¹¹ Notably, a Cs/FA/MA triple-cation system stabilized the α -phase of FAPbI₃,

achieving a record-low bandgap of 1.48 eV. Processing innovations also demonstrated bandgap control capabilities: systematic studies revealed that annealing temperatures influence MAPbI₃ grain size and bandgap, with high-temperature treatments reducing defects at the expense of bandgap widening.¹² Vapor-assisted crystallization methods further enabled precise bandgap engineering through halogen partial pressure regulation.^{13,14} Collectively, these advancements established a comprehensive toolkit for perovskite bandgap manipulation.

However, these developments introduced new scientific controversies and unresolved challenges. While cation engineering enables bandgap tuning, studies cautioned that Cs⁺ concentrations exceeding 10% may induce lattice distortions.¹⁵ Parallel concerns emerged

regarding lead-free alternatives (Sn/Ge/Bi-based systems), where toxicity concerns (e.g., Sn^{4+}) and efficiency limitations persist.¹⁶ A particularly complex issue-PIPS-gained prominence following Hoke et al.'s 2015 hyperspectral imaging study, which visualized nanoscale halide (Br^-/I^-) migration under illumination.⁵ Subsequent systematic analyses by Slotcavage et al. attributed this phenomenon to halogen ion migration and localized band bending, resulting in phase-separated low-bandgap domains.¹⁷ These findings catalyzed extensive research into mitigating phase segregation while maintaining bandgap tunability.

3. PHOTOINDUCED PHASE SEGREGATION PHENOMENON IN MIXED HALIDE PEROVSKITE MATERIALS

3.1. Basic Concepts of Photoinduced Phase Segregation

The earliest research on PIPS can be traced back to 2013, when mixed halide perovskite materials gained significant attention due to their tunable bandgap and high photoelectric conversion efficiency. Despite the large optical bandgap of mixed halide perovskites, the V_{OC} did not increase as expected. Hook et al.⁵ observed a red shift in the photoluminescence (PL) spectrum under illumination, with a new red-shifted peak appearing at 1.68 eV in the PL spectrum of $(\text{CH}_3\text{NH}_3)\text{Pb}(\text{Br}_{1-x}\text{I}_x)_3$. This was accompanied by an increase in sub-bandgap absorption (around 1.7 eV), indicating the formation of luminescent trap states, as shown in Figure 2a,2b. X-ray diffraction (XRD) analysis revealed that illumination caused the splitting of XRD peaks, and these changes were completely reversed after a few minutes in the dark. The researchers speculated that photoexcitation might lead to the formation of iodine-rich and bromine-rich regions, with the former acting as recombination center traps. This instability could limit the maximum voltage achievable in mixed halide PSCs. The researchers emphasized the importance of further studies, particularly in suppressing photoinduced defects, to improve device performance and stability. Subsequent reports also confirmed that PIPS indeed affects device performance.¹⁸

In the initial stages following the identification of the PIPS, research efforts were predominantly confined to foundational aspects, characterized by preliminary exploratory investigations. Early studies on suppressing this phenomenon primarily focused on empirical mitigation strategies. For example, Xiao et al.¹⁹ proposed a method to suppress halogen redistribution by coating the surface of nanoscale grains with self-assembled long-chain organic ammonium cations (such as *n*-butylammonium). Through testing, they found that the V_{OC} could be linearly tuned from 1.05 to 1.45 V, with a maximum external quantum efficiency of 5%. This study successfully suppressed halogen redistribution in mixed halide perovskites through surface coating. Concurrently, parallel efforts in mechanistic investigations emerged, attempting to unravel the underlying causes of this phenomenon through fundamental theoretical analyses. Balakrishna et al.²⁰ prepared perovskite films with varying ratios of bromide and iodide and treated them with iodide. Their results demonstrated positive effects on phase segregation, dark recovery, and photoelectric performance. Early studies also investigated the mechanisms of PIPS. Samu et al.²¹ studied the impact of long-term illumination on the performance of mixed halide lead PSCs. Their results showed that long-term illumination affected both photovoltaic performance and external quantum efficiency. The researchers proposed mechanisms involving charge recombination centers and energy gradients to explain these observations.

3.1.1. Phase Segregation under Illumination and Phase Recovery in the Dark. Studies indicate that photogenerated charges interact with the surrounding atomic lattice, creating localized excited states. These interactions cause local deformation of the semiconductor atomic bonds, resulting in strain. Under low-intensity illumination, the number of polarons generated is small, leading to a local strain gradient. This strain gradient causes I^- and Br^- to separate within the lattice, forming iodine-rich and bromine-rich regions. Therefore, in the absence of light, all strain gradients disappear, and an entropy-driven ion remixing process occurs immediately.^{22,23} In

the study by Cho et al.,²⁴ it was found that the absorption peak at 460 nm increased with recovery time. The researchers conducted experiments with a temperature gradient, monitoring the bleaching maximum at 460 nm at different temperatures and performing kinetic fitting. They found that the rate constant of dark recovery followed the Arrhenius relationship, with an activation energy of 41.9 ± 1.8 kJ/mol. They concluded that the dark recovery process is driven by entropy-driven halogen ion remixing, and that increasing temperature accelerates this process. Studies have also shown that the crystal orientation remains unchanged throughout the process of separation and dark-state recovery, indicating that ion rearrangement occurs within the original lattice framework rather than through the recrystallization of amorphous domains to form new nuclei.²⁵

Further research has found that, Lin et al.²⁶ found that excess free charges could accelerate ion segregation in perovskite materials, thereby reducing material stability, as shown in Figure 2c,2d. Therefore, effective carrier extraction could significantly improve the photostability of PSCs. After experimental validation, the researchers proposed that as illumination proceeds, the injection of a large number of photogenerated carriers creates a nonuniform distribution of carriers within the material. The electric field generated by this nonuniform distribution of carriers could induce internal strain, lowering the energy barrier for ion migration and creating an electric field effect. The electric field caused by carrier distribution could weaken the interactions between ions, promoting the generation of more structural defects and accelerating ion migration. Excess free charges could shield and weaken the ionic and hydrogen bonds within the perovskite lattice, making ion migration easier. Similarly, excess charges could neutralize ion vacancies, reducing the binding force caused by Coulombic interactions and thus promoting ion migration. Similar views were also proposed by Bischak et al.²⁷ and Barker et al.²⁸ In a study by Belisle et al.²⁹ on halide segregation in perovskite materials, the researchers found that carriers were trapped at surface states, creating an electric field directed toward the film surface. Halide vacancies migrated and accumulated near the surface states to shield the trapped electric field. This migration led to local changes in chemical composition, promoting the formation of photoinduced traps.

3.1.2. Phase Segregation and Recovery under Continuous Illumination. As research on PIPS continued, scientists discovered that perovskite materials that had undergone PIPS could not only recover to their initial state in the dark but also undergo similar recovery under continuous high-intensity illumination. In an experiment by Mao et al.,³⁰ a pulsed laser (400 nm) was used to irradiate single-crystal $\text{MAPb}(\text{Br}_{0.8}\text{I}_{0.2})_3$ microplates at laser intensities of 10 W/cm², 200 W/cm², and then back to 10 W/cm². The results showed that under low-intensity illumination (10 W/cm²), the emission peak gradually red-shifted from around 550 nm to around 710 nm over 60 s, indicating the formation of iodine-rich regions. Under high-intensity illumination (200 W/cm²), three emission peaks were immediately observed at 545, 610, and 710 nm. Over time, the peaks at 545 and 610 nm red-shifted and blue-shifted, respectively, eventually merging into a single peak at around 555 nm, while the peak at 710 nm gradually disappeared. When low-intensity illumination was reapplied, the peak at 555 nm red-shifted again, returning to the initial phase segregation emission. A similar experimental phenomenon was observed in the study by Kuno et al.,²³ where the researchers noted that under high-intensity illumination, the number of strain-induced carriers increased significantly, and the strain was uniformly distributed throughout the lattice, eliminating the strain gradient. As a result, there was no driving force for ion separation, leading to the remixing of I^- and Br^- .

The above experiments only observed phase segregation and recovery under continuous illumination, but in the experiment by Guo et al.,³¹ further progress was made. In this experiment, the researchers first conducted gradient experiments with light intensity and illumination time, concluding that PIPS and photoinduced self-healing are common in mixed halide perovskite films. In subsequent experiments, the researchers observed that continuous high-intensity illumination caused visible cracks on the film surface, as shown in

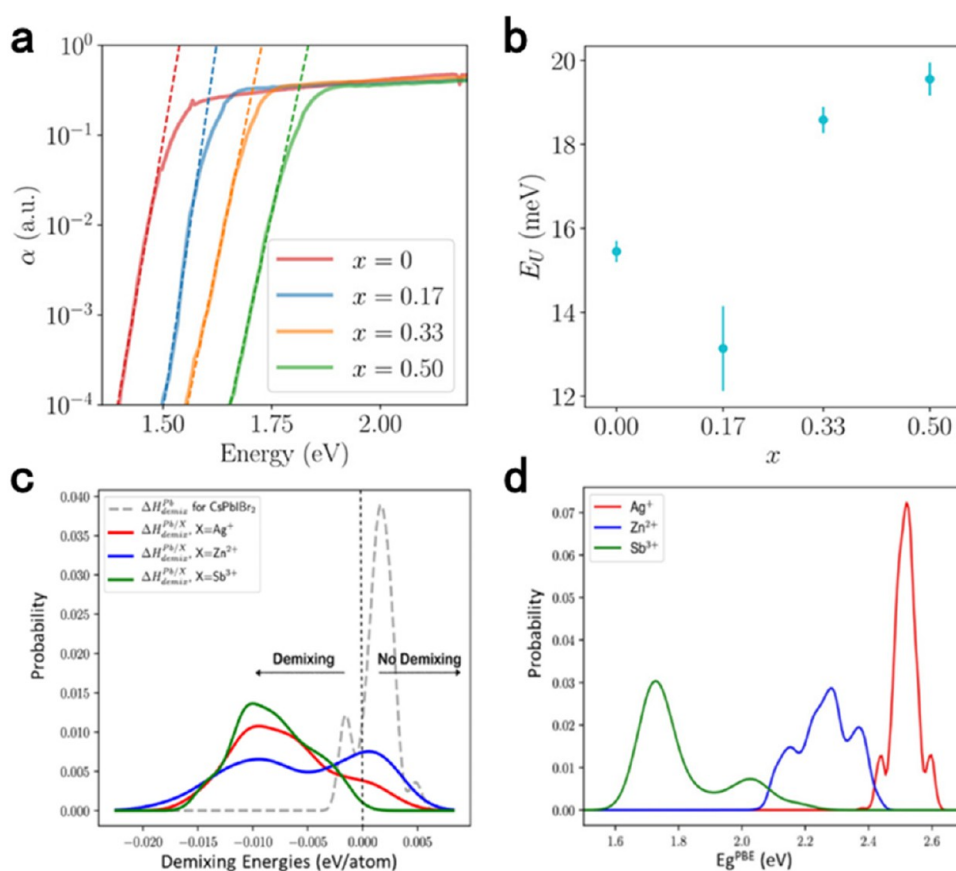


Figure 3. Analysis based on thermodynamic model: (a) Absorption coefficients of perovskite films with different bromide contents measured by Photothermal Deflection Spectroscopy (PDS).³³ (b) Urbach energies of perovskite films with different bromide contents.³³ Reproduced from ref 33. Copyright 2023 American Chemical Society. (c) Distribution of demixing enthalpy changes across different systems.³⁴ (d) Electronic bandgap energy distribution calculated from 1000 molecular dynamics trajectory frames at 300 K based on the PBE level of theory.³⁴ Reproduced from ref 34. Copyright 2022 American Chemical Society.

Figure 2e. Cracks are a direct result of the lattice's inability to withstand accumulated strain, that is, mechanical fractures that occur after exceeding the strain tolerance threshold, and this phenomenon also explains the nonrepeatability of polycrystalline thin film PIHS. Cracks eliminate the gradient driving halide ion migration by releasing strain, preventing phase separation of the material after cycling. Through follow-up experiments, they successfully verified the reversibility of halogen ion migration in perovskite films before reaching the limit and the irreversibility after exceeding the limit, proving that the self-healing phenomenon in perovskite films under high-intensity light has a limit. Based on this research trajectory, future investigations in this field may establish a more comprehensive model grounded in polaron theory to elucidate the fundamental principles underlying this threshold. Additionally, promising research directions could include identifying factors governing stress accumulation/release, exploring whether this threshold can be modulated via doping or practical fabrication techniques, and determining whether quantification and characterization methods can achieve equilibrium between peak performance and stress thresholds.

4. METHODS FOR ANALYSIS PHASE SEGREGATION

4.1. Thermodynamic Models

In thermodynamic modeling, the free energy-as a state function-plays a pivotal role in describing the energy availability of systems under specific conditions and determining process directionality and equilibrium criteria. Its application facilitates the simplification of complex system analyses, unifies macroscopic and microscopic descriptions, and exhibits universal applicability across disciplines.

For instance, the Urbach energy quantifies the degree of static and dynamic disorder in materials. Perovskites with enhanced disorder typically demonstrate superior optoelectronic conversion efficiency and operational stability.⁵² Phase segregation phenomena in perovskites can also be evaluated through their intrinsic disorder levels, which profoundly influence band tail states. These tail states arise from deviations in the ideal periodic crystal arrangement due to defects or doping, leading to abrupt transitions in conduction and valence bands rather than idealized distributions. Specifically, Urbach energy characterizes the exponential decay of absorption spectra below the band edge, reflecting material optical properties. Consequently, Urbach energy directly governs photovoltaic device performance: higher Urbach energy correlates with reduced V_{OC} and carrier mobility, thereby degrading overall efficiency. Studies utilizing Urbach energy analysis (Figure 3a,b) reveal that static disorder contributes less significantly to Urbach energy in perovskites compared to other materials like InGa_N systems, attributed to the smaller effective mass of charge carriers-a parameter describing quasi-particle behavior during crystalline transport. Subsequent investigations demonstrate that short-range potential fluctuations become negligible in perovskites due to this small effective mass, while experimentally measured Urbach energies consistently exceed computational predictions. This discrepancy highlights the dominant role of temperature-dependent dynamic disorder over static alloying-induced disorder in perovskite materials.³³ Given the intrinsic connection between PIPS and material disorder, such thermodynamic modeling provides critical insights into phase segregation mechanisms-a representative application of thermodynamic frameworks in this context. Moreover, the systematic correlation analysis between

Urbach energy characteristics and experimental outcomes exemplifies a widely adopted methodology in this field.

Thermodynamic models elucidate PIPS by incorporating light-induced modifications to free energy landscapes. Photogenerated carriers alter the free energy system through carrier-lattice interactions and chemical potential redistribution. The spatial separation of excitons under illumination induces lattice distortions, increasing disorder and lowering activation energy for phase segregation. Concurrently, nonequilibrium component distributions under illumination disrupt thermodynamic equilibrium. These models enable quantitative analysis of energy variations pre- and postillumination, facilitating predictions of threshold conditions (e.g., light intensity, compositional gradients) and segregation pathways. Additionally, thermodynamic frameworks quantify the driving forces in phase segregation, where entropy (system disorder) and enthalpy (internal energy) jointly govern the process—increased entropy and decreased enthalpy synergistically drive phase separation. Combining Monte Carlo simulations and density functional theory (DFT) calculations, researchers map ion migration potential energy surfaces to predict irreversible phase segregation under illumination. Such models further enable coupled opto-thermo-electric field analysis in PSCs, leading to modified phase diagrams. For example, when investigating dopant effects on suppressing phase segregation in CsPbIBr₂ under illumination (Figure 3c,3d), enthalpy change calculations revealed Sb³⁺-doped systems as most segregation-prone, while Ag⁺ and Zn²⁺ doping enhanced stability.³⁴ Complementary studies employ Arrhenius plots to analyze thermodynamic barriers, identifying optimal chlorine doping ratios in trihalide systems through activation energy variations.³⁵ DFT-based mixing energy simulations further demonstrate that low-concentration Cl[−] substitutions stabilize I[−] sites and induce vertical crystal growth. This mechanism elucidates Cl-mediated stabilization in low-dimensional perovskites, providing theoretical guidance for fabricating efficient wide-bandgap low-dimensional PSCs.³⁶

In summary, in the study of phase separation induced by light, the common research ideas of thermodynamic models mainly include evaluating the influence of modification operations on stability through thermodynamic parameters and establishing thermodynamic models based on the principle of minimum energy, simulating and calculating the core indicators in the preparation process, and conducting relevant experiments based on this theoretical basis, ultimately achieving the expected experimental results. Future research needs to further unify the quantitative standards for thermodynamic parameters (such as clarifying enthalpy change types), and explore in depth the impact of nonequilibrium thermodynamics (such as the dynamic evolution of ΔG under photoexcitation) on device performance.

4.2. Molecular Dynamics Models

Molecular dynamics (MD) simulations, a computational approach rooted in classical Newtonian mechanics, investigate the temporal evolution and interactions of atomic/molecular systems. This method provides dynamic insights into structural evolution beyond static configurations, enabling statistical mechanical correlations—such as thermodynamic properties, kinetic parameters, and ensemble-averaged behaviors—via trajectory analysis. In PIPS research, MD applications span multiple scales. For instance, crystallization kinetics during thin-film fabrication influence grain boundary defect density and phase purity, directly impacting phase stability. Techniques like grazing-incidence X-ray diffraction (GI-XRD) dynamically monitor annealing-induced crystal growth, offering actionable insights for process optimization.³⁷ Charge carrier dynamics, including defect-mediated nonradiative recombination at interfaces, are recognized contributors to PIPS. Ultrafast spectroscopic techniques and electrochemical impedance spectroscopy probe time-resolved carrier transport and recombination dynamics. While direct observation of halide ion migration underlies empirical PIPS studies, MD simulations quantitatively model ion migration pathways and activation energies under varying temperatures or electric fields.

For example, a multidimensional kinetic analysis of halogen ion oxidation during perovskite phase segregation identified three driving forces: oxidation potential gradients, photochemical potential differences, and dark-state-induced potential disparities. Here, photochemical potential differences are also discussed in Section 3.1.1, serving as further evidence and an extension of the PIPS phenomenon driven by carrier migration. This work finally dual migration mechanisms for oxidized halogen species: vapor-phase diffusion to adjacent regions and solid-phase diffusion through lattice defects (e.g., vacancies).³⁸ This shows how MD-integrated kinetic models elucidate migration pathways and mechanistic hierarchies in PIPS.

At interfaces, chemical corrosion between perovskites and charge transport layers critically affects device longevity. MD simulations analyze carrier distribution and lifetime at these interfaces. Transient absorption spectroscopy studies demonstrated that chlorine-doped films exhibit prolonged carrier lifetimes compared to undoped counterparts. Subsequent MD-based investigations linked this enhancement to suppressed photoinduced halide segregation under continuous illumination in doped systems.³⁵ Advanced studies employ first-order kinetic models to fit spectroscopic data, correlating excitation intensity and temperature dependencies with phase segregation kinetics. This study finally find out phase segregation rate constants increase with excitation intensity until reaching a threshold; elevated temperatures accelerate both segregation and dark-state recovery rates.²⁴ These kinetic analyses unveil halogen ion migration behaviors and quantitative parameters during PIPS and recovery, advancing mechanistic understanding and guiding stable PSC design.

Further theoretical frameworks integrate MD-derived insights into device engineering. For instance, an empirical model correlating PL-observed halide segregation amplitudes with trap-mediated carrier recombination ratios and absorbed photon flux was developed to probe surface trap state-PIPS relationships. The impact of carrier recombination on PIPS, as mentioned here, exhibits similarities with the description of excess free charges on PIPS introduced in Section 3.1.1. The more photons absorbed here, the higher carrier concentration, and consequently the more pronounced phase segregation becomes. This model provides predictive capabilities for optimizing interfacial passivation strategies.

$$I = f(gGt) = A \frac{(gGt)^2}{(1 + (gGt)^2)} (1 - e^{-BgGt}) \quad (1)$$

In this model, G denotes the photogenerated carrier generation rate per unit volume ($\text{cm}^{-3} \text{ s}^{-1}$), t represents time (s), A and B are dimensionless global fitting parameters, and g is a local fitting parameter describing the halide segregation propensity induced by a single photogenerated carrier within a unit volume. This empirical framework not only rationalizes halide segregation kinetics across varying illumination conditions but also provides guidelines for suppressing segregation by minimizing trap-mediated carrier recombination rates, thereby informing the design of stable PSCs.³⁹

Simulating molecular motion allows MD to investigate the properties and behaviors of materials. This approach is particularly valuable in PIPS studies, enabling atomic-level insights into intermolecular interactions, carrier transport trajectories, and crystallization pathways.⁴⁰ Such simulations deepen the mechanistic understanding of perovskite synthesis, offering critical references for controlling PIPS and optimizing fabrication protocols. Furthermore, advancements in artificial intelligence have fostered integration between MD and machine learning. In practical machine learning applications, neural network (NN) algorithms can be employed to predict perovskite compositions that simultaneously exhibit a high tolerance factor and maintain a wide bandgap. Such compositions enable the reduction of bromine content while preserving device performance, thereby mitigating the risk of bromine segregation. Experimentally, these predicted compositions have demonstrated a smaller photoinduced red-shift, indicating enhanced photostability.⁴¹ However, based on the existing understanding of PIPS and the principles of applying machine learning in this field, a more

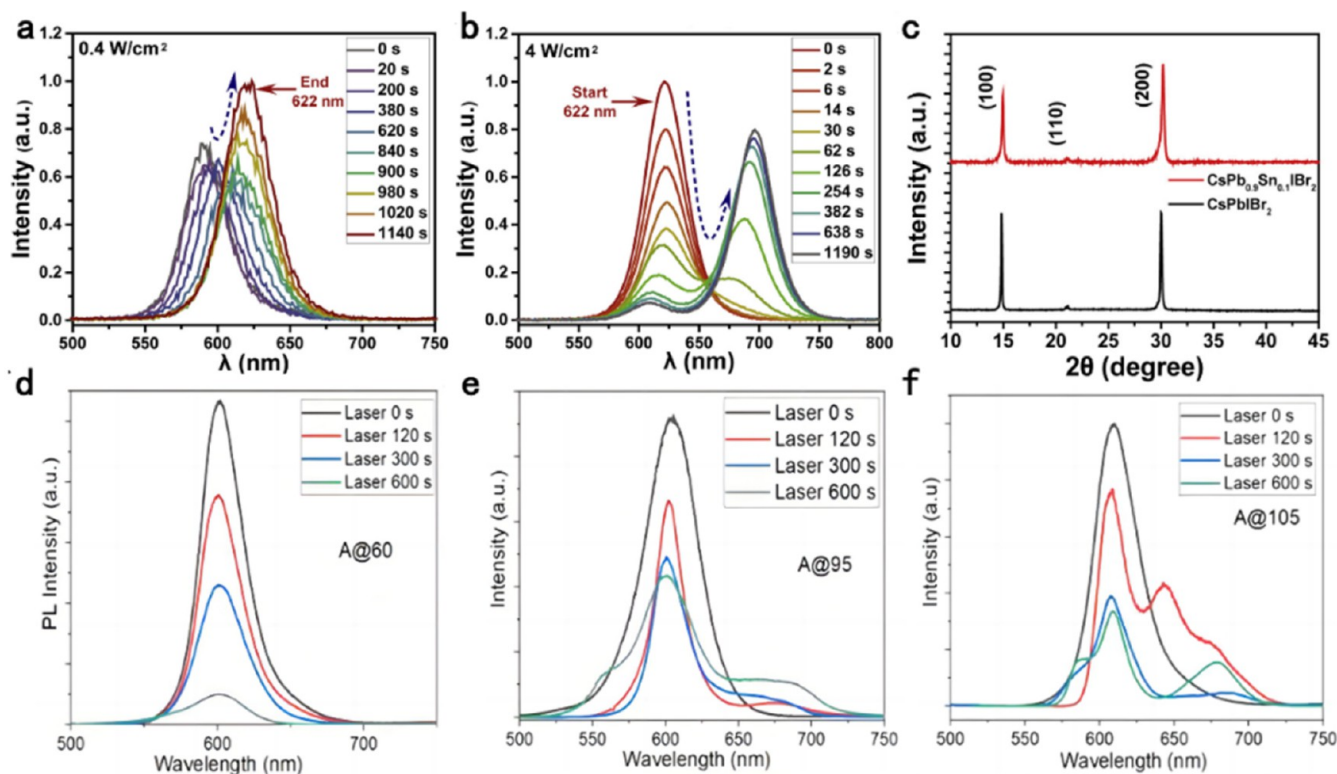


Figure 4. Effect of grain boundary strain modulation on photoinduced phase segregation behavior: (a) PL variation of CsPbIBr₂ thin film at excitation power density of 0.4 W/cm².⁴⁷ (b) PL variation of CsPbIBr₂ thin film under excitation power density of 4 W/cm².⁴⁷ (c) XRD patterns of CsPb_{0.9}Sn_{0.1}IBr₂ thin film (red solid line) and CsPbIBr₂ thin film (black solid line).⁴⁷ Reproduced with permission from ref 47. Copyright 2020 Elsevier. (d–f). Normalized PL emission spectra.⁵⁰ Reproduced from ref 50. Copyright 2021 American Chemical Society.

comprehensive approach may be developed. By incorporating key factors such as the ratio of X-site halogen atoms, surface defect states, and band alignment, along with secondary factors including grain boundary dimensions and nonradiative recombination rates, as training features for machine learning models, it may become possible to achieve accurate prediction of PIPS behavior in the future. Large-scale data-driven MD models now enable predictive material design by computationally screening perovskite compositions and architectures for superior optoelectronic performance, thereby guiding experimental synthesis efforts.⁴²

4.3. Latest Research Progress in Grain Boundary Strain Models

Grain boundaries (GBs) refer to the interfacial regions between crystalline grains within polycrystalline perovskite films, and its strain arises from lattice mismatch between adjacent grains or stress accumulation during growth, leading to atomic misalignment and localized stress fields. Studies on GB-PIPS correlations reveal that phase segregation dynamics are influenced by GB strain and compositional gradients. For instance, films with varying Br[−] content exhibit distinct phase segregation extents under identical stress, attributed to Br-dependent strain modulation at GBs.⁴³ In mixed-halide PSCs, GB strain weakens lattice confinement of halide ions, reducing their migration activation energy and enabling light-enhanced ion diffusion along GBs. Concurrently, strain-induced lattice distortions create “ion highways” that accelerate directional halide migration, thereby promoting PIPS. Supporting this, *J*–*V* hysteresis studies demonstrate that forward-to-short-circuit scans yield higher efficiencies compared to reverse scans, ascribed to iodine-rich GB regions elevating mobile ion densities and interfacial strain, which impedes charge extraction.¹⁸ A “lattice anharmonicity” model further posits that charge funneling in iodine-rich zones enhances lattice anharmonicity, inducing strain that facilitates photoactivated ion displacement—a mechanism linking GB strain to carrier transport anomalies and PIPS amplification.⁴⁴ Additionally, strained GBs

generate vacancies and interstitial defects, acting as halide trapping centers. Photogenerated carrier recombination at these defects releases energy that drives ion migration, while defect-induced local electric fields polarize migration pathways, spatially exacerbating PIPS. Therefore, providing high-purity and highly crystalline nucleation sources has been reported to fundamentally reduce the number and activity of adverse grain boundaries, thereby achieving superior environmental stability.⁴⁵

Recent advances focus on strain engineering to counteract PIPS. Research has found that due to the blocking effect of GBs on carrier transport, carriers trapped in iodine-rich terminal domains cannot be output because of grain boundary obstruction, thus exacerbating nonradiative recombination and ultimately intensifying performance loss. However, for vertically oriented crystals, crystals aligned along the charge transport direction can provide vertical conductive paths. Therefore, even with more defects and more significant segregation, carriers can still be efficiently output to achieve performance stability.⁴⁶ Meanwhile, Sn²⁺ doping at the B-site induces lattice contraction, as evidenced by reduced lattice constants in XRD (Figure 4a–4c). PL spectroscopy under continuous illumination reveals enhanced intensity and red-shifted emission (576 → 676 nm), indicating stabilized intermediate phases. Mn²⁺ doping similarly reduces lattice strain via contraction.⁴⁷ Kim et al.⁴⁸ employed ethylenediammonium diiodide additives to simultaneously refine grain size and introduce hollow structures, experimentally validating strain mitigation. Lattice contraction—marked by decreased unit cell volume, shortened lattice spacing, and reduced defect density—directly alleviates GB strain by minimizing defect-driven stress. This contraction also lowers lattice energy, enhancing structural stability and reducing internal stress. Furthermore, the application of in-plane stress to induce confined grain growth, thereby transforming grain boundaries from open to compact structures, has also been demonstrated to suppress the occurrence of phase segregation.⁴⁹

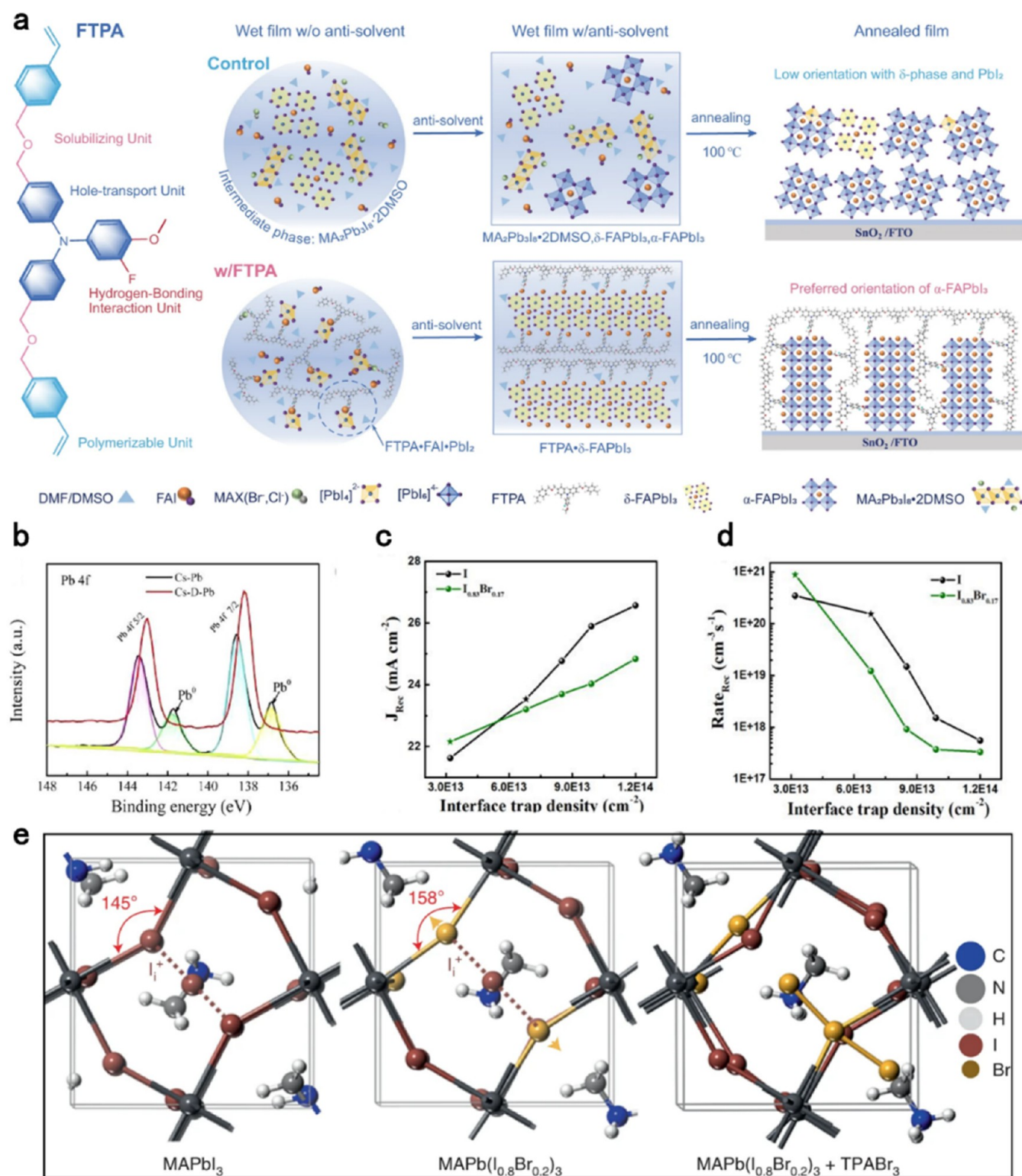


Figure 5. Implementation of surface state doping strategy: (a) The core mechanism of FTPA.⁵³ Reproduced from ref 53. Available under a [terms of the Creative Commons CC BY] license. Copyright 2023 Springer Nature. (b). XPS spectra shows the electron density around the Pb atom increases.⁵⁴ Reproduced from ref 54. Copyright 2024 American Chemical Society (c). Composite current density.⁵⁵ (d). The relationship between the radiative recombination rate of perovskite/HTL interface and the interface well density of I and $\text{I}_{0.83}\text{Br}_{0.17}$ devices under V_{OC} .⁵⁵ Reproduced from ref 55. Copyright 2020 American Chemical Society. (e). Geometric Structure of Iodine Gap (I^+) in MAPbI_3 and $\text{MAPb}(\text{I}_{0.83}\text{Br}_{0.17})_3$.⁶² Reproduced with permission from ref 62. Copyright 2023 Springer Nature.

Hu et al.⁵⁰ identified a GB size threshold governing PIPS in perovskite films. By annealing temperature modulation, they fabricated films with graded grain sizes. XRD and PL analyses

revealed phase stability for grains <43 nm but pronounced segregation at ≥ 43 nm (Figure 4d–4f). Corresponding PSC metrics (V_{OC} , short-circuit current density, fill factor, power conversion efficiency)

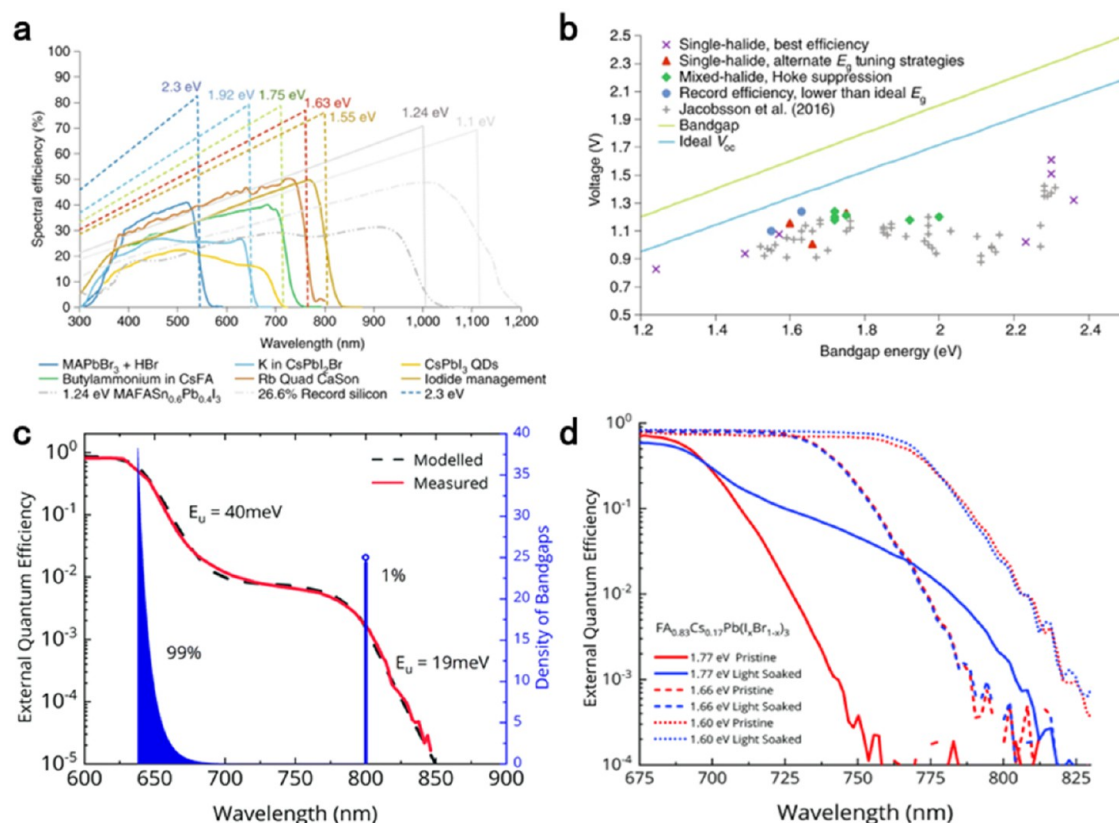


Figure 6. Impact of photoInduced phase segregation on the performance parameters of solar cell devices: (a) Spectral efficiency of various potential perovskite candidate ions.⁶⁵ (b) V_{OC} of PSCs with different bandgaps.⁶⁵ Reproduced with permission from ref 65. Copyright 2018 Springer Nature. (c) The external quantum efficiency for photovoltaic devices spectra after the phase separation of the pristine and halides, as well as the distribution of the band gap in the perovskite materials.⁶⁴ (d) The changes in the External Quantum Efficiency for Photovoltaic Devices spectra of $\text{FA}_{0.83}\text{Cs}_{0.17}\text{Pb}(\text{I}_{1-x}\text{Br}_x)_3$ solar cells with different band gaps before and after illumination are shown.⁶⁴ Reproduced with permission from ref 64. Copyright 2019 Royal Society of Chemistry.

mirrored this trend, with optimal performance below the threshold. Mechanistically, smaller grains confine carriers, preserving intrinsic luminescence and stability, whereas larger grains enable light-driven halide redistribution into iodine-/bromine-rich domains that act as recombination centers.

In the research on lattice strain and PIPS, the viewpoint that PIPS is influenced by GB strain and corresponding composition was first proposed. Subsequently, related studies focused on how to control phase separation phenomenon by controlling grain boundary size and achieved certain results, proving that reducing grain boundary size can suppress photo induced phase separation phenomenon to a certain extent. Subsequently, new research in this field focused on exploring the correlation between GB size and the intensity of PIPS, and ultimately determined a GB threshold for the perovskite material used in this study. Following this thinking logic, it will be found that there are still some loopholes in this research direction. First, no verification experiments were conducted using perovskite materials with different compositions, and the existence and distribution of “GB thresholds” for different materials, as well as the influence of different material compositions on the final results, were not explored. Meanwhile, there is no clear report on whether the lattice treatment of perovskite materials will affect other functional layers during device operation. These may be the possible future focuses of this research direction.

4.4. Latest Research Progress in Surface State Doping Models

Surface-state doping, a critical strategy in PSCs engineering, modulates interfacial electronic structures, chemical bonding, and defect distributions at perovskite surfaces or GBs, effectively suppressing PIPS. Surface passivation via dopants involves chemical bonding at defect sites, filling undercoordinated vacancies, and

reducing ion mobility to hinder halide migration. As previously discussed, the thermodynamic driving force for PIPS originates from Gibbs free energy differences between halide phases. Surface-state doping alters surface energy to influence phase separation tendencies. Dopants may also form electron transport layers (ETLs) at perovskite surfaces, enhancing photoelectron extraction and suppressing non-radiative recombination, thereby mitigating PIPS.^{51,52}

A common surface engineering approach targets the inhibition of intermediate phase formation. For instance, Li et al.⁵³ developed a surface modifier, FTPA, which suppresses complex intermediate phases during perovskite crystallization. As evidenced by XRD, SEM, and UV–vis analyses during spin-coating and annealing (Figure 5a), FTPA promotes oriented crystallization, improves film quality, and enhances device stability. Similarly, DDBA, another surface modifier, strengthens Pb–N coordination via XPS-confirmed electron density enhancement around Pb atoms (Figure 5b), passivating under-coordinated Pb²⁺ and eliminating metallic Pb signals linked to degradation and recombination.⁵⁴ Concurrently, Br doping suppresses PbI₂ formation, as shown by XRD (Figure 5c,d), reducing interfacial trap density while maintaining phase stability.⁵⁵ Intermediate phases, often arising from precursor solution interactions, disrupt crystal alignment and carrier transport. Tailored surface-state doping strategies, guided by compositional and structural characterization, effectively inhibit such phases. However, excessive Br doping introduces trade-offs: moderate Br incorporation reduces defect density and boosts V_{OC} in wide-bandgap PSCs, but overdoping increases deep-level traps via Pb–Br–Pb bond angle expansion and iodine interstitial defect formation (Figure 5e), exacerbating non-radiative recombination and V_{OC} losses. Similar studies have also been reported, such as adding THB to form bonds between sulfur atoms

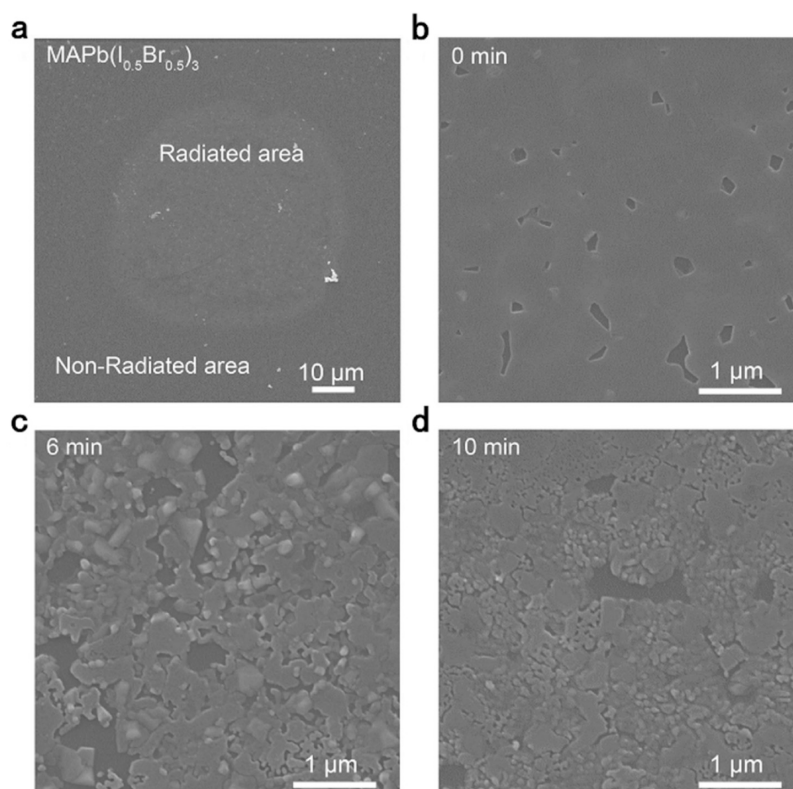


Figure 7. Microstructural observations of morphology degradation in perovskite films leading to phase segregation: (a–d). High-magnification SEM images of MAPb(I_{0.5}Br_{0.5})₃ thin films after irradiation at varying durations.⁶⁸ Reproduced from ref 68. Available under a [terms of the Creative Commons CC BY] license. Copyright 2021 Wiley-VCH.

and the undercoordinated lead ions on the perovskite surface, and introducing potassium ions to passivate halogen vacancies via bonding.^{56,57}

Surface doping also leverages ion-specific bonding to passivate defects. For example, PDA dopants utilize diammonium groups to bond with Pb–X clusters at GBs, neutralizing surface defects.⁵⁸ Formate (HCOO[−]) ions suppress halogen vacancies by forming hydrogen bonds with undercoordinated Pb²⁺. To address residual MAI-induced Cs⁺ aggregation, FAAC additives employ “intermediate-phase crystallization,” where acetate (Ac[−]) forms hydrogen bonds with formamidinium (FA⁺), enhancing surface crystallinity and efficiency.⁵⁹ Guo et al.⁶⁰ introduced AIDCN, an organic additive with moderately reactive amino groups (e.g., primary amines), which inhibit iodide migration via hydrogen bonding with I[−], as experimentally validated.

Current surface-state strategies predominantly focus on chemical passivation, ion doping, and ligand engineering, often exhibiting passive and singular functionalities. Future innovations may explore active and multifunctional approaches. For example, developing surface state dopants containing dynamic bonds such as disulfide bonds to achieve automatic defect repair under illumination or heating conditions; Develop hydrophobic passivation integrated materials such as fluorinated polymers, while providing water oxygen barrier and defect passivation.⁶¹ Meanwhile, it is also possible to consider combining porous materials such as MOFs to limit lattice expansion and suppress ion migration.

5. INFLUENCE OF PHOTOINDUCED PHASE SEGREGATION PHENOMENON ON DEVICE PERFORMANCE

Since the rise of solar cells as a typical representative of new energy applications, PSCs have attracted significant attention from researchers due to their advantages such as high efficiency, low cost, simple fabrication processes, and broad

spectral absorption range. However, PIPS not only affects the optoelectronic conversion properties of perovskite thin films but also becomes a critical factor influencing the performance and stability of solar cell devices. This section will delve into the mechanisms of PIPS in PSC devices and explore how to scientifically regulate this phenomenon to effectively enhance the performance and stability of PSCs, laying a more solid foundation for the further development of PSC technology.

5.1. The Influence Mechanism of Photoinduced Phase Segregation on Solar Cell Devices

PIPS multifaceted impacts on perovskite solar cell devices, with the most immediate consequences observed in their core optoelectronic performance. The fundamental operation of these devices relies on efficient carrier transport; however, studies indicate that under illumination, interactions between electron and hole transport layers may occur, where phase-segregated regions act as barriers to charge transport, thereby reducing charge collection efficiency.⁶³ Concurrently, reports suggest that phase segregation induces photon accumulation within single-phase domains, severely compromising photon utilization. The resultant reduction in light absorption contributes to short-circuit current losses. As illustrated in Figure 6, research has demonstrated that phase segregation in mixed-halide perovskite layers significantly degrades device performance:^{64,65} bandgap variations caused by phase segregation lead to energy level mismatches that reduce V_{OC} , while interfacial transport obstruction further deteriorates fill factor.⁶⁵

Beyond core photovoltaic performance, PIPS critically impacts essential optoelectronic parameters such as photoelectric conversion efficiency and operational stability. Chen et

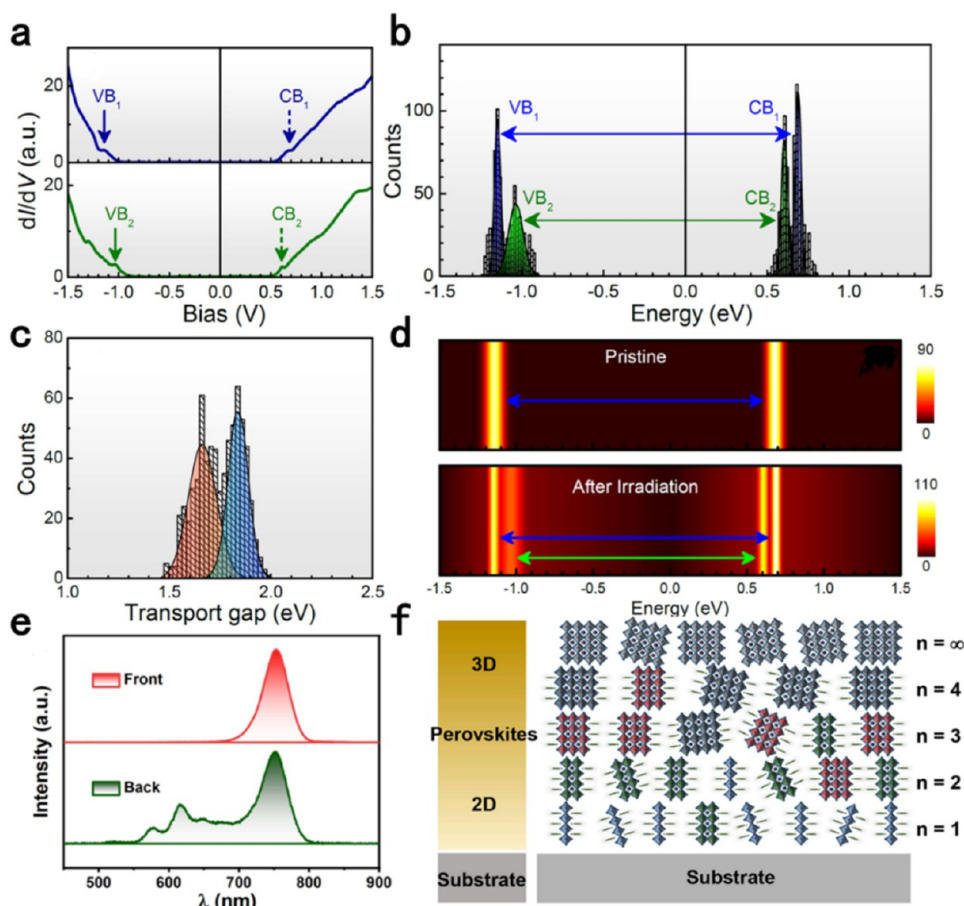


Figure 8. Distribution characteristics of dual-band formation and spatially inhomogeneous phase segregation within the device: (a) Representative dI/dV spectra.⁷⁰ (b) C_B (conduction band) and C_V (valence band) histograms.⁷⁰ (c) Transport gap of the sample after 15 min of illumination.⁷⁰ (d) Band edge position comparison before and after illumination using chromaticity diagrams.⁷⁰ Reproduced from ref 70. Copyright 2023 American Chemical Society. (e) PL spectra of the front and back sides of a perovskite film under 405 nm laser excitation.⁷¹ (f) Schematic illustration of distinct 2D phase formation in quasi-2D perovskite thin films.⁷¹ Reproduced from ref 71. Copyright 2023 American Chemical Society.

al.⁶⁶ demonstrated that phase segregation-induced bandgap variations in the perovskite layer hinder effective absorption of specific light wavelengths, thereby reducing power conversion efficiency. Expanding beyond photovoltaic effects, research has increasingly focused on structural implications of phase segregation. Leijtens et al.⁶⁵ identified that light-induced halide redistribution under illumination introduces photoinstability, while thermally accelerated phase segregation at elevated temperatures exacerbates structural degradation, leading to thermally induced instability. Azhar et al.⁶⁷ corroborated these findings, observing that internal structural modifications from phase segregation amplify instability factors such as increased grain boundary defects. As shown in Figure 7, Li et al.⁶⁸ reported morphological defects including voids and island-like structures under similar experimental conditions. These studies collectively confirm that phase segregation fundamentally compromises both structural integrity and photovoltaic performance.

Emerging investigations reveal extended consequences. Phase segregation degrades material and interfacial quality, manifesting as high-density trap states in the absorption layer and poor-quality perovskite/charge transport layer heterojunctions that exacerbate nonradiative losses. Notably, ETLs exhibit pronounced photoluminescence (PL) quenching.⁶⁴ While transient suppression of nonradiative recombination has

been attributed to band alignment between iodine-rich and bromine-rich domains, prolonged illumination ultimately increases recombination channels through dynamic formation and dissolution of these segregated regions.⁶⁸ This progression reduces luminescent efficiency and overall device stability. Liu et al.⁶³ and Sheng et al.⁶⁹ further linked phase segregation to photocurrent hysteresis, where uneven charge transfer between segregated phases induces current–voltage curve hysteresis, undermining device reliability and consistency.

Recent studies by Paramanik⁷⁰ reveal that phase segregation induces dual-band formation in perovskite solar cell devices. As illustrated in Figure 8a–8d, PIPS creates iodine-rich domains with narrower bandgaps and bromine-rich domains with wider bandgaps. This bandgap dichotomy introduces energy barriers in carrier transport pathways, while concomitant energy level offsets modify carrier transport dynamics. Concurrently, grain boundaries have been identified as active sites for iodide migration and accumulation, localizing iodine-rich regions near boundaries and amplifying material inhomogeneity. These structural irregularities impose additional energy obstacles for carrier transport, degrading overall charge transport performance.

Xiaorong Qi et al.⁷¹ further demonstrated spatially heterogeneous effects of phase segregation across device architectures. As shown in Figure 8e,8f, upper layers—

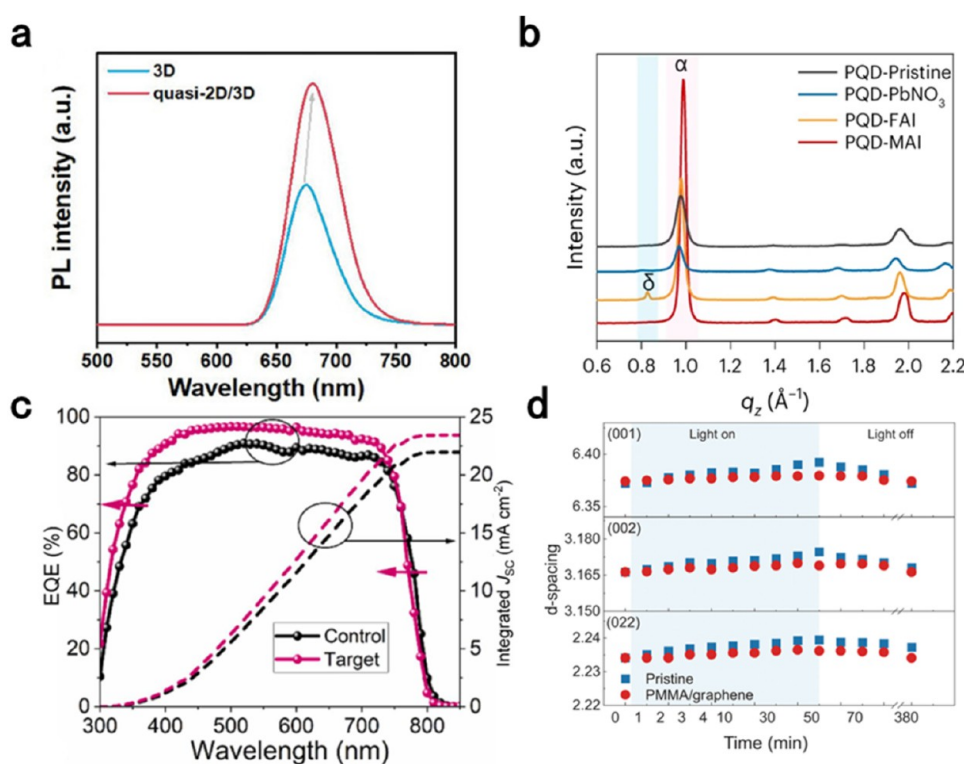


Figure 9. Schematic of device-level phase segregation suppression strategies (a). PL spectra of 3D and quasi-2D/3D perovskite thin films.⁷³ Reproduced with permission from ref 73. Copyright 2025 John Wiley and Sons. (b). PQD wide-angle X-ray scattering results after different ligand exchange methods.⁷⁶ Reproduced with permission from ref 76. Copyright 2024 Springer Nature. (c). EQE Spectroscopy and Total Integrated J_{sc} of control sample (unmodified by Cs-based perovskite recrystallized-arrays) and target sample (modified by Cs-based perovskite recrystallized-arrays).⁸⁴ Reproduced with permission from ref 84. Copyright 2023 Elsevier. (d). Changes in d values of different reflective surfaces obtained through XRD analysis.⁸³ Reproduced with permission from ref 83. Copyright 2025 The American Association for the Advancement of Science.

particularly those containing large- n phase regions resembling 3D structures—exhibit pronounced phase segregation and instability, which may trigger cascading segregation events and further performance degradation. In contrast, small- n phase configurations in lower layers suppress phase segregation and enhance photostability. Leveraging this spatial differentiation, the research group developed optimized 2D perovskite designs for lower device regions, achieving improved stability and enhanced operational performance.

5.2. Suppression of Photoinduced Phase Segregation in Solar Cell Devices

The impacts of PIPS occurring in solar cell devices can be broadly categorized into two types. The first involves effects within individual functional layers, such as carrier redistribution, ion migration, and defect generation, leading to the degradation of photoelectrical efficiency parameters and the formation of structural defects within each layer. The second type pertains to effects at the interfaces between functional layers, exemplified by poor-quality perovskite/charge-transport layer heterojunctions or the formation of interfacial energy barriers. Correspondingly, mainstream strategies to address these phenomena in devices include structural optimization, collaborative design between the perovskite and charge transport layers, and optical management measures aimed at mitigating damage induced by external stimuli from excess ultraviolet light.

5.2.1. Structure Optimization. To address the detrimental effects of light-induced phase segregation on device performance, researchers have developed various structural

modification strategies. A prominent approach involves the construction of 2D/3D heterostructures, where two-dimensional perovskite layers are integrated with conventional three-dimensional counterparts. This design capitalizes on the inherent advantages of 2D materials to mitigate stability issues caused by uncoordinated ions at surfaces and grain boundaries of traditional 3D perovskites.⁷² For instance, the introduction of short-chain diammonium iodide (PDADI) creates Dion-Jacobson phase quasi-2D perovskites with optimized n -values. As demonstrated in Figure 9a, such 2D layers effectively absorb ultraviolet photons (<400 nm), thereby reducing lattice damage from high-energy radiation while stabilizing carrier concentration fluctuations.⁷³ Notably, quasi-2D architectures have demonstrated remarkable operational stability under continuous illumination, maintaining over 90% of initial power conversion efficiency after 500 h of maximum power point tracking.⁷⁴ The n -value is a core parameter in the design of two-dimensional perovskite materials, which can be balanced, bandgap width, carrier mobility, and stability by adjusting the n -value. There are also related reports indicating that when the n value is small (usually 1 or 2), the content of organic interstitial cations is higher, further inhibiting light induced halogen separation.⁷¹ In the study of two-dimensional structures, there are also reports linking it to additive engineering. Related reports indicate that by adding chlorine to the two-dimensional perovskite precursor solution, vertical crystal growth can be induced, thereby improving the performance and stability of wide bandgap PSCs.³⁶

For instance, based on a mixed-halide perovskite system, spin-coating a PEABr solution to introduce a two-dimensional

Ruddlesden–Popper (RP) phase onto the three-dimensional perovskite can optimize band alignment and reduce interfacial energy loss. Additionally, this approach enhances resistance to moisture and oxygen ingress. Simultaneously, the passivation effect provided by PEABr significantly improves the open-circuit voltage. Through the synergistic combination of this structured surface treatment and effective passivation, the overall device performance is enhanced, while the occurrence of phase segregation is suppressed.⁷⁵ These studies illustrate that adopting a “composition-structure” strategy during experimentation often not only allows the doped ions to exert their effects but also benefits from the advantages brought by their own multidimensional structures, leading to more outstanding outcomes. This treatment also reduces perovskite defect states and enhances photoelectronic stability, suggesting that when applied to mixed-halide perovskites, it can suppress PIPS to some extent.

Quantum dot architectures (0D structures) have emerged as another promising structural solution. These nanoscale systems (<100 nm) exhibit quantum confinement effects, high surface-to-volume ratios, and restricted carrier transport. Methylammonium iodide (MAI) ligand engineering has proven effective in improving quantum dot stability, as shown in Figure 9b, where MAI-treated quantum dots demonstrate suppressed δ -phase formation and reduced phase segregation.⁷⁶ It has also been reported that the introduction of CuInSe₂ quantum dots can significantly suppress ion migration channels within grains, but has little effect on ion migration at grain boundaries.⁷⁷ However, inhibiting ion migration can also to some extent suppress the occurrence of photo induced phase separation phenomenon. It is worth mentioning that quantum dot structures not only play a role in maintaining stability, but also reduce surface trap density,⁷⁶ improve optoelectronic performance,⁷⁸ and have a wide absorption range and narrow emission peak.⁷⁶ Therefore, the heat of zero dimensional quantum dot structures is gradually increasing. In addressing the narrow absorption range of all-inorganic solar cells, relevant research has demonstrated that constructing a CsPbI₂Br/PBDTTT-E-T:IEICO integrated heterojunction solar cell extends the EQE response from 680 to 900 nm and significantly enhances both the short-circuit current density and fill factor.⁷⁹

5.2.2. Optical Management Systems. In terms of phase stability research at the device level, optical management systems have gradually been applied as a phase stability strategy in recent years. Researchers have found that the high energy of ultraviolet light (<400 nm) directly damages the perovskite lattice and intensifies the migration of halide ions;^{80,81} Non uniform light absorption can lead to high local carrier concentration, causing lattice stress accumulation and ion migration;⁸² The heat generated by light will accelerate ion migration and phase separation.⁸³ Based on similar reports mentioned above, it is not difficult to see that an effective light management system plays a crucial role in phase stability. After researchers discovered some degree of side effects of ultraviolet light, the focus of the study shifted to how to add structures similar to filter layers to eliminate the influence of ultraviolet light. Recent studies have demonstrated that Cs-based perovskite recrystallized arrays exhibit a blue-shift trend in external quantum efficiency (EQE) (Figure 9c), indicating enhanced absorption efficiency in short-wavelength regions such as ultraviolet light. Moreover, testing confirmed superior stability.⁸⁴ Other reports have employed single-crystal TiO₂

particles to optimize ETLs, noting that oxygen vacancies on TiO₂ surfaces undergo photoinduced desorption under Ultraviolet Illumination, leading to rapid trap-assisted recombination. In contrast, devices with SCs-based architectures show blue-shifted absorption spectra, reducing high-energy UV photon absorption and thereby improving photostability. Experimental results also indicated better stability.⁸⁵ Additionally, the introduction of SCN[−] has been shown to induce a slight blue shift in film absorption edges, increasing bandgap energy and enhancing phase stability.⁸⁶ These findings collectively suggest that suppressing short-wavelength light effects (<400 nm) effectively mitigates phase segregation.

To address inhomogeneous light absorption, researchers frequently employ surface nanostructures—such as antireflective coatings, photonic crystals, or bioinspired arrays—to minimize reflection losses and homogenize optical field distribution, thereby reducing localized photothermal effects and ion migration driving forces. For instance, antireflective coatings (e.g., MgF₂) and plasmonic strategies have been reported to enhance light-harvesting efficiency by reducing reflection losses.⁸⁷ Another study proposed utilizing triboelectric fields to actively regulate carrier distribution, balancing p–n junction barrier heights and suppressing ion migration and lattice strain caused by carrier accumulation.⁸² These approaches provide novel insights for alleviating phase instability induced by uneven light absorption.

Regarding photothermal management, one strategy involves precisely modulating depletion zone electric fields within perovskite layers to counteract light-induced ion migration driving forces, thereby achieving superior photothermal stability.⁸⁸ Building on earlier discussions of dynamic stress mechanisms under illumination, a recent study integrated monolayer graphene onto perovskite surfaces via poly(methyl methacrylate) (PMMA) polymer interfacial coupling. As shown in Figure 9d, the PMMA/graphene interface stabilized lattice spacing, leveraging graphene's exceptional mechanical properties and polymer coupling effects to double the modulus and hardness of perovskite films while significantly suppressing light-induced lattice expansion/contraction.⁸³

Current photomanagement systems mitigate phase separation by regulating light absorption distribution, minimizing high-energy photon damage, and suppressing photothermal effects, primarily through proactive interventions. Future directions may involve developing adaptive optical structures (e.g., liquid crystal dimming layers) responsive to outdoor light intensity fluctuations. Additionally, integrating multiphysical field designs—combining optical, thermal, and electric field synergies—could further optimize performance. For instance, electric fields could guide ion migration paths while photomanagement reduces migration driving forces. Cost-effective large-area photomanagement technologies should also be prioritized for industrial scalability.

5.2.3. Synergistic Interactions. Beyond perovskite layer optimization, recent research has expanded to full-device engineering, emphasizing synergistic interactions between perovskite, electron transport, and hole transport layers. In conventional wide-bandgap PSCs, SCN[−] incorporation inhibits PIPS by blocking halogen ion migration pathways and elevating activation energy.⁸⁶ At the perovskite/C60 interface, reactive passivators that form low-dimensional phases with lead halides have been shown to suppress nonradiative recombination while maintaining phase stability.⁸⁹ A bilateral passivation strategy using coadsorbed (4-(3,6-dimethyl-9H-carbazol-9-

yl)butyl)phosphonic acid (Me-4PACz) and phthalocyanine on NiO_x substrates improved interfacial morphology and contact, simultaneously suppressing nonradiative recombination and preserving phase stability.⁹⁰

A summary of the latest research contents discussed above is provided in Table 1. While current interfacial passivation innovations effectively suppress phase separation and enhance efficiency/stability, challenges remain in durability, environmental compatibility, and scalable manufacturing. Future research should prioritize time-dependent passivation efficacy, eco-friendly materials, and industry-oriented designs to advance commercialization of mixed-halide.

6. CONCLUSION AND OUTLOOK

Since the emergence of perovskite materials as next-generation optoelectronic materials, mixed-halide perovskites have gained widespread attention as semiconductor materials due to their tunable bandgap and excellent energy level matching. However, in solar cell applications particularly, PIPS has significantly compromised device performance. This review systematically examines research progress on PIPS and summarizes mitigation strategies from the perspective of solar cell device engineering.

This review categorizes four dominant mechanistic frameworks for phase segregation. Thermodynamic and kinetic models primarily serve as analytical tools, employing classical parameters (e.g., energy variation, entropy/enthalpy changes) to assess system stability through thermodynamic equilibrium analysis and kinetic simulations. These models also guide material design by optimizing compositions and structures through energy minimization principles. The grain boundary strain model correlates phase segregation with lattice strain, with recent research focusing on strain engineering through ion doping and annealing protocols. Gradient lattice engineering has identified critical lattice parameters for phase stability, establishing structure–property relationships. Current challenges lie in systematically mapping strain thresholds across compositional variations. Surface state doping strategies emphasize defect passivation via chemical bonding and suppression of intermediate phases, achieving improved crystal quality. While effective, these approaches remain largely reactive; future directions may involve proactive material design integrating multiple functionalities like hydrophobicity-passivation hybrids.

Regarding device impacts, this work elucidates how phase segregation affects key photovoltaic parameters: V_{OC} reduction through nonradiative recombination pathways, short-circuit current loss via accumulation of photon, and fill factor deterioration from increased series resistance. The cumulative effects manifest as efficiency losses and stability issues. Current device-level stabilization strategies include dimensional engineering (2D/3D heterostructures and quantum dot 0D configurations) to suppress ion migration, and photon management systems filtering long-wavelength irradiation to minimize lattice distortion. Emerging research explores interfacial interactions between perovskite and charge transport layers, though detailed mechanistic understanding remains limited. Future investigations may focus on interlayer engineering for synergistic phase stabilization across functional layers.

The emergence of phase segregation induces systemic degradation across the entire system, exemplified by the formation of energy barriers, intensification of GB strain, and

Table 1. Summary of the Primary Strategies Currently Employed to Suppress PIPS in Mixed-Halide Perovskites

strategy category	experimental approach	mechanism of action	key impact on device performance
Structure Optimization	Construction of 2D/3D Heterostructures ⁷³	Leverages complementary advantages of 2D and 3D structures to coordinately address ionic disorder at surfaces and grain boundaries. ⁷³	Reducing lattice damage from high-energy radiation while stabilizing carrier concentration fluctuations. ⁷³
	Application of Quantum-Dot Structures ^{76,77}	Utilizes quantum confinement effects, high surface-area-to-volume ratio, and confined carrier transport properties to suppress ion migration and reduce phase segregation effects. ^{76,77}	Maintains stability while reducing surface trap density, enhancing optoelectronic performance, and offering broad absorption range with narrow emission peaks. ^{76–78}
Optical Management Systems	Implementation of UV-Filter-Like Layer Structures ^{84–86}	Introduces specific structures that induce a blue shift, thereby mitigating the impact of ultraviolet light. ^{84–86}	Increases bandgap energy and enhances phase stability, while effectively alleviating phase segregation. ^{84–86}
	Use of Anti-Reflection Coating-Like Structures ⁸⁷	Reduces reflection loss and homogenizes the optical field distribution, thereby diminishing local photothermal effects and the driving force for ion migration. ⁸⁷	Leads to more uniform carrier distribution, mitigating ion migration and lattice strain caused by carrier accumulation. ⁸⁷
Synergistic Interactions	Introduction of Reactive Passivators ^{86,89}	Forms a low-dimensional phase with lead halides at the perovskite/C60 interface. ^{86,89}	Suppresses nonradiative recombination while maintaining phase stability. ^{86,89}
	Dual-Side Passivation Strategy ⁹⁰	Improves interface morphology and contact conditions. ⁹⁰	Inhibits nonradiative recombination and preserves phase stability. ⁹⁰

exacerbation of defect states. As shown in Figure 10, such system-wide degradation further diminishes the overall

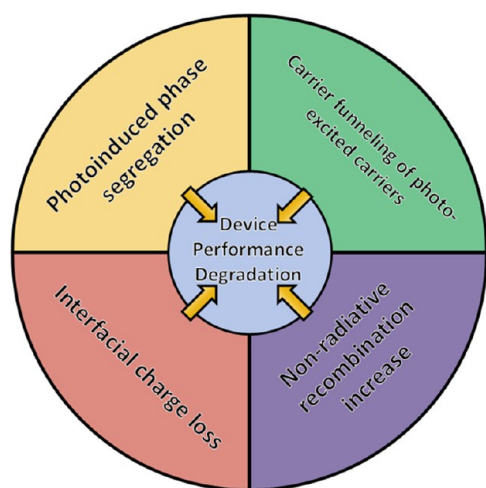


Figure 10. Causal relationships between PIPS and its derived phenomena.

performance of the solar cell through aggravated nonradiative recombination, most notably reflected in a reduction of the V_{OC} . Concurrently, these structural alterations may establish a microenvironment more favorable for ion segregation, thereby allowing the phase separation process to become partially self-sustaining or even self-accelerating once initially triggered.

For future research directions, at the mechanistic exploration level, developing multidimensional characterization models—such as in situ techniques to track the spatiotemporal evolution of strain-defect-recombination dynamics—to resolve the interplay between phase segregation and device degradation. At the material innovation level, designing wide-bandgap phase-segregation-resistant perovskites (e.g., via Sn^{2+} doping) to intrinsically suppress halide migration while maintaining optoelectronic performance. At the Device engineering level, Exploring synergistic interface passivation and stress engineering strategies, such as combining 2D capping layers with grain boundary passivation, to simultaneously mitigate interfacial recombination and lattice strain accumulation. This integrated approach—spanning atomic-scale defect control, mesoscale phase stabilization, and macroscale device optimization—will pave the way for phase-stable, high-efficiency mixed-halide PSCs.

■ ASSOCIATED CONTENT

Data Availability Statement

Data will be made available on request.

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Notes

The authors declare no competing financial interest.

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